Output-only experiments and recovery before Hamiltonian calibration

Astra research construction

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Completed capability. Capture a first arbitrary, unencoded many-body quantum input through calibrated coherent noise outputs; learn its previously unknown local Hamiltonian from later outputs of the same source; then recover that original input and its external reference correlations. No later copy of the original input and no direct source preparation, control or readout is required.

New construction. Retained replacement-state purifiers supply local process probes. A signed emission record has a fixed-order mixed derivative equal to one Hamiltonian coefficient. Finite interpolation acquires that derivative with all failed attempts counted. An adaptive-decoder lemma permits calibration records to remain correlated with the archived quantum state. A bounded-neighborhood companion removes equilibration waits.

Governing scope. Actual pure, calibrated, losslessly collected Weyl ports, a complete local Hamiltonian class, stationarity, protected receiver memory and paid finite controls are substantive premises. The known-Hamiltonian continuous reader is the accepted P1 operation at revision `324acb589443002bf8d529cc15e160129c5e36d2`. Calibration works at every $r > 0$; generic first-payload recovery uses P1's sufficient condition $r > \beta_0$.

Limits. No competitive laboratory implementation, fixed-hardware asymptotic reader, arbitrary unknown-port calibration, natural-system completeness or historical priority is established. Explicit conservative constants are extremely large. The completed consequence is a finite polynomial-resource capability theorem.

The previous full classical-feedback acquisition theorem, separate global-budget result, matched fixed-order parity/GHZ law, finite-slew and meter calibrations, and thermal-effusion operation are preserved unchanged.

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1 Question, consequence and strongest alternatives

The next question is whether knowing the apparatus dynamics *before the first specimen arrives* is essential for quantum acquisition. In the specified interface, it is not. The receiver can acquire an undecoded quantum record, use future emissions to learn the dynamics, and decode the first record later. The target is the original quantum channel, including arbitrary references; a classical state estimate would not achieve it.

The operative principle is that a pure replacement channel exports two resources: the preceding source state and a purifier of its replacement. The first carries the payload; the second makes subsequent dynamics experimentally accessible. Discarding the second can erase every stationary Hamiltonian-identification signal in the first.

Three results are separate: a heralded local process oracle at any $r > 0$; an unconditioned correlation experiment learning a fixed-locality Hamiltonian class with inverse-square accuracy dependence up to logarithms; and a first-payload recovery corollary removing prior knowledge of H. The finite receiver implementation is inherited from P1, with its growing demands charged.

Stationary-output identification is established: Guţă–Kiukas identify ergodic quantum Markov dynamics up to phase and unitary coordinate changes, and derive local asymptotic normality [4]. Their general result does not itself give this uniform many-body, local-hardware, first-payload recovery construction. No new claim that outputs contain dynamical information is made.

Efficient derivative-based Hamiltonian and Lindbladian learning is established [5]; high-temperature Gibbs and real-time learning have strong sample and computational guarantees [7]. Pseudo-Choi methods can be substantially more efficient when their controlled forward/inverse evolution access is supplied [6]. Here the required correlation is physically acquired from a source accessible only through its outgoing calibrated fields.

Gong et al.'s October 2026 *Out-of-control Hamiltonian Learning* reconstructs generic two-local models using uniform source preparation and measurements, and specified lattice models using computational-basis preparation and measurements [8]. Those are restrictive but still direct source interfaces. This construction supplies its probes through outgoing purifiers.

Liu et al.'s October 2026 local-probe construction learns bounded-degree two-body models with one measurable qubit, fast single-qubit source controls and a reference backbone, achieving Heisenberg precision scaling [9]. That is a strong alternative when those controls exist; no precision or runtime superiority over it is claimed. Navascués constructs probabilistic resetting of uncontrolled systems with interacting probes and unknown Hamiltonians [10]. Its access and generic-success promises differ; its general $O(d_S^3)$-probe construction is exponential in source qubit count.

Relative to accepted P1 [1], the achieved new consequence is precise: H need not be supplied before collecting the first arbitrary payload. The contribution is the finite polynomial local acquisition and uniform channel composition, beyond a qualitative combination of generic output identifiability and a known-model decoder. Port axes, phases, rates and coherent delivery remain supplied. No separation from every possible same-output-access calibration algorithm, or historical novelty of the complete composition, has been established.

2 Premises and the inherited acquisition primitive

Let S contain n qubits, $d = 2^n$, with static Hamiltonian

$$H = \sum_{a=1}^{M} \theta_a P_a, \quad |\operatorname{supp} P_a| \le k, \quad B_H \ge \sum_a |\theta_a|.$$

The distinct nonidentity Pauli list is complete and known, with unknown coefficients. Taking all weight-at-most-k Paulis allows unknown support and gives $M = O(n^k)$ at fixed k. An identity

term is irrelevant and unidentifiable. Define

$$\mathcal{L}_H(\rho) = -i[H,\rho] + r\sum_i(\mathcal{E}_i(\rho)-\rho), \quad \mathcal{E}_i(\rho) = I_i/2 \otimes \mathrm{Tr}_i\,\rho, \quad \Lambda = 2B_H + 2nr.$$

Known nonidentity Weyl ports $L_{i,\mu} = \sqrt{r}\,W_{i,\mu}/2$, pure vacuum inputs and actual lossless coherent output access are premises. They fix the physical Pauli frame. The physical photon rate $R_{\mathrm{ph}} = 3nr/4$ differs from the synthetic reset rate r.

P1's finite-bin receiver supplies

$$V|\psi\rangle = \sqrt{1-p}\,|\psi\rangle_S|\emptyset\rangle + \sqrt{p}\,|\Phi_2\rangle_{S,A}|\psi\rangle_B, \qquad p = 1-\mathrm{e}^{-r\Delta}. \tag{1}$$

Flags are orthogonal with input-independent probabilities. B contains the old state with all references; A purifies the fresh source. This receiver operation grants no source SWAP or controlled source reset. P1 already uses replacement purifiers in its diagonal companion; the primitive itself is inherited.

For supplied $\beta(H) \le \beta_0 < r$, where

$$\beta(H) = \max_i \sum_{X\ni i}(|X|-1)\,\mathrm{osc}(h_X), \qquad \kappa_0 = r-\beta_0,$$

P1's reverse-history receiver obeys

$$\left\|\overline{\mathcal{D}}_{H,T}\mathcal{C}_{H,T} - \mathrm{id}\right\|_\diamond \le 2\sqrt{n\mathrm{e}^{-\kappa_0 T}}. \tag{2}$$

Its collector needs known noise parameters and norm bounds, not the coefficients of H. Only the eventual inverse-history decoder needs those coefficients. This distinction permits deferred calibration.

3 A local process oracle from outgoing fields

Temporarily start the source in I/d. For X of size b, retain each site's *last* replacement purifier in a front window of width w. Let the source evolve for τ, discarding intervening output. In a back window of width w, retain the *first* export from each site of X. Accept when all required resets occur; its ideal probability is $(1-\mathrm{e}^{-rw})^{2b}$.

If H is omitted only during the two windows, the accepted state is the normalized Choi state of

$$\mathcal{T}_X^\tau(\rho_X) = \mathrm{Tr}_{\bar X}\,\mathrm{e}^{\tau\mathcal{L}_H}(\rho_X \otimes I_{\bar X}/2^{n-b}).$$

Restoring H during those windows changes the state by at most $4B_Hw$ on every ideal reset record: telescope unitary segments of total duration $2w$ through normalized conditional CPTP reset maps. No inverse clock probability multiplies this timing bound.

This gives local process tomography. Bell-projecting a receiver input with the retained purifier, accepting the identity Bell outcome, adds success probability 4^{-b} and yields a chosen-input query to $\mathcal{T}_X^\tau$, reference-uniformly. Failed queries consume the input; there is no repeat-until-success claim on one unknown state. Choi tomography and teleportation are established machinery; the construction is their acquisition from these actual ports.

The $4B_Hw$ estimate is a state bound. Writing $\xi = 4B_Hw < 4^{-b}$, a conservative input/reference-uniform bound after the Bell postselection is $2\xi/(4^{-b}-\xi)$. An unconditional physical instrument error must also be divided by a lower bound on the full accepted probability. Thus window width, clock successes and Bell conditioning are all paid. The main learning construction below retains failures as outcomes and needs no such division.

4 The signed-output identity

For candidate $P = P_a$, choose $i \in \operatorname{supp} P$ and single-site Pauli Q_i anticommuting with P_i. Put $R_P = iPQ_i$, a signed Hermitian Pauli, $b = |\operatorname{supp} P|$, and $s = b + 1$. Use the front window on $\operatorname{supp} P$, but only one back export at i. Measure R_P^T on the front purifiers and Q_i on that export. The raw result is their signed product on success and zero otherwise:

$$Y \in \{-1, 0, 1\}, \quad f_P(w, \tau) = \mathbb{E}Y, \quad \mathbb{E}Y^2 = (1 - \mathrm{e}^{-rw})^s.$$

Every attempted window is counted. Purifiers can instead be measured immediately in their fixed Pauli bases, retaining only the last classical outcome per site. Deferred measurement gives the same law, so this calibration needs no long-lived entangled register.

Theorem 1 (Acquired mixed derivative)**.** *For source start I/d,*

$$\boxed{\partial_w^s \partial_\tau f_P(0, 0) = 2s!\, r^s \theta_P.} \tag{3}$$

Proof. Success needs b selected front resets and one selected back reset. At order w^s, these are the only window insertions. Any Hamiltonian action, extra reset or waiting expansion there adds a power of w. The front resets create Bell pairs and the back reset exports the required output, so

$$[w^s] f_P(w, \tau) = r^s 2^{-n} \operatorname{Tr}[R_P \mathrm{e}^{\tau \mathcal{L}_H^\dagger}(Q_i)].$$

Its value at $\tau = 0$ is zero, while its derivative divided by r^s is

$$2^{-n} \operatorname{Tr}[R_P(i[H, Q_i] - rQ_i)] = 2\theta_P.$$

Pauli orthogonality removes every other coefficient and the known depolarizing term. Differentiating w^s supplies $s!$. □

The derivative is an identity, not an assumed experimental primitive.

5 Finite interpolation, all-trial sampling and precision

The front protocol has a finite CPTP record representation: overwrite each selected purifier at every reset and set a seen bit. The back protocol freezes its first selected export. Including these constant-size registers, each window generator has diamond norm at most Λ, as does the middle generator. Consequently

$$f_P(w, \tau) = \sum_{u,v \geq 0} c_{uv} w^u \tau^v, \qquad |c_{uv}| \leq \frac{(2\Lambda)^u \Lambda^v}{u!\, v!}. \tag{4}$$

These matrices are an analysis device: the algorithm never builds a 2^n-dimensional matrix. The source generates the observations.

One explicit, deliberately loose schedule sets

$$h = (8\Lambda)^{-1}, \qquad W_{D,s} = 4(D + 1)^2 (2D^2/h)^{s+1}.$$

Choose $D \geq \max(s, 2)$ with

$$8 W_{D,s} 4^{-D} \leq \eta s! r^s. \tag{5}$$

Use tensor Chebyshev–Lobatto nodes on $[0, h]^2$, differentiate the degree-D interpolant at $(0, 0)$, and divide by $2s!r^s$.

To verify the bias, truncate the Taylor series at degree D in each variable. This rectangle retains the exact desired derivative. Equation (4) bounds its nodal remainder by $4\,4^{-D}$. DCT-I interpolation amplifies the supremum norm by at most $2(D + 1)$ per variable. Repeated

Markov polynomial inequalities give the data-to-derivative bound $W_{D,s}$. Equation (5) makes the coefficient bias at most $\eta/4$.

Estimate every nontrivial nodal mean within

$$\zeta = \frac{\eta s! r^s}{2W_{D,s}}, \qquad p_{\max} = (rh)^s.$$

Since $\mathrm{Var}(Y) \le p_{\max}$, a sufficient Bernstein count is

$$N_{\mathrm{node}} = \left\lceil 8\left(\frac{p_{\max}}{\zeta^2} + \frac{1}{\zeta}\right)\log\frac{2M(D+1)^2}{\delta}\right\rceil. \tag{6}$$

Jointly, all statistical coefficient contributions are at most $\eta/4$ except with probability δ. Nodes with $w = 0$ are known exactly zero and need no source trial. All other attempts, including zeros, are paid.

Round durations *inside* $[0, h]$. Because $|\partial_w f| \le 2\Lambda$ and $|\partial_\tau f| \le \Lambda$, timing accuracy $\zeta/(12\Lambda)$ changes a mean by at most $\zeta/4$. Compute the final coefficient weights with certified total ℓ_1 error at most $\eta/8$. Bias, sampling, timing and arithmetic then fit within coefficient error η. Exact algebraic clock values are not granted.

At fixed k, D is logarithmic up to logarithms of logarithms in the dimensionless size and precision parameters. The full raw-trial count is

$$N_{\mathrm{cal}} = \widetilde{O}_k\left[M\left(\frac{\Lambda}{\eta}\right)^2\left(\frac{\Lambda}{r}\right)^{k+1}\right]. \tag{7}$$

The tilde hides a high power of logarithms: one direct bound has D^{4k+14} before the confidence logarithm. Equations (5)–(6), not an optimistic constant, govern this claim. Sharper factorial tails and exact derivative-weight norms are legitimate improvements but are not silently substituted in the resource evaluation below.

A sharper implemented resource option uses the actual computable coefficient-weight norm $A_D = \sum_j |a_j|$ and the factorial tail

$$R_D = \frac{2\mathrm{e}^{3/8}4^{-(D+1)}}{(D+1)!}.$$

Choose D with $A_D R_D \le \eta/4$, after a certified allowance for weight rounding. The randomized-node procedure below also works after burn-in and removes the separate nodal-batch factor. This is an available stronger schedule, not merely a suggested optimization; its resource arithmetic is reported alongside the deliberately loose schedule.

6 Paid source preparation and the no-burn-in companion

6.1 General fixed locality

The preceding identity requires a maximally mixed exterior, which must be acquired. Discard output for a prescribed burn-in time.

Lemma 2 (Complete source replacement)**.** *For every joint source/reference state,*

$$\left\|(\mathrm{e}^{t\mathcal{L}_H} \otimes \mathrm{id}_R)(\rho_{SR}) - I/d \otimes \rho_R\right\|_1 \le 2\sqrt{n\log 2}\,\mathrm{e}^{-rt/2}.$$

Proof. Define $D_t = D(\rho_{SR}(t)\|I/d \otimes \rho_R) = n\log 2 - S(S|R)$. The source Hamiltonian does not change it. Reset i has entropy production

$$D(\rho\|\mathcal{E}_i\rho) + D(\mathcal{E}_i\rho\|\rho) \ge \log 2 - S(S_i|S_{\bar{i}}R).$$

Strong subadditivity and the entropy chain rule give $\sum_i S(S_i|S_{\bar{i}}R) \le S(S|R)$. Hence $\dot{D}_t \le -rD_t$, $D_0 \le 2n\log 2$, and Pinsker proves the result. Singular states follow by continuity. □

With a fixed trial cap, choose

$$t_{\mathrm{burn}} = \frac{2}{r} \log \frac{2\sqrt{n \log 2}\, N_{\mathrm{cal}}}{\epsilon_{\mathrm{mix}}}$$

before every trial. Telescoping makes the whole calibration protocol ϵ_{mix}-close to exact replacement-start trials. This is paid exposure. It proves neither an efficient all-rate decoder nor a conditional reset-history decoupling claim.

6.2 Bounded interaction neighborhoods

A stronger start result has a stronger locality premise. Let C_i contain every candidate support touching i, and suppose $\max_i |C_i| = c$ is fixed. Front-reset all of C_i, using identity observables on purifier sites outside $\operatorname{supp} P$. Put $s = |C_i| + 1$. The leading reset term replaces C_i and traces its previous correlations. Every term in $[H, Q_i]$ lies in C_i. Thus (3) holds for *every* starting state, including one entangled with an exterior and archived payload; the analytic bounds are uniform.

Fixed interpolation nodes on different uncontrolled source states cannot be combined by assuming cancellation. Freshly randomize the node each trial. Let a_j be the final rational coefficient weights, including division by $2s!r^s$, and $A = \sum_j |a_j|$. Choose dyadic probabilities $q_j \geq |a_j|/(2A)$, with a zero-output dummy branch for the remainder. For J nodes, choose $2^\ell \geq 4J$ and

$$q_j = 2^{-\ell} \left\lceil 2^{\ell-1} |a_j|/A \right\rceil .$$

This sums to at most $3/4$ and uses ℓ fair bits. Pad selection latency to a fixed duration.

Return $Z = a_j Y / q_j$. Conditional on all past records, its mean is the interpolation functional of *one current-state function*. Its bias is therefore uniformly bounded, while

$$|Z| \leq 2A, \qquad \mathbb{E}[Z^2 \mid \mathrm{past}] \leq 2A^2 (rh)^s.$$

A martingale Bernstein/Freedman bound gives

$$N = O\left[\left(\frac{A^2 (rh)^s}{\eta^2} + \frac{A}{\eta} \right) \log \frac{M}{\delta} \right]$$

raw trials per coefficient. There is no equilibration wait or independent-source-state assumption. H and the ports must remain stationary. The exponent depends on c, not merely k; growing neighborhoods are not hidden in a fixed-k claim.

7 Deferred calibration and the first arbitrary payload

Lemma 3 (Adaptive decoder with correlated calibration)**.** *Let σ_z be the subnormalized archived-bank/reference state conditional on later record z, with unchanged unconditioned marginal. Suppose every good record obeys $\left\| \mathcal{D}_{\widehat{H}(z)} - \mathcal{D}_H \right\|_\diamond \leq e$, and the total bad probability is at most δ, uniformly over initial inputs. Then*

$$\left\| \sum_z (\mathcal{D}_{\widehat{H}(z)} \otimes \mathrm{id})(\sigma_z) - (\mathcal{D}_H \otimes \mathrm{id})(\sum_z \sigma_z) \right\|_1 \leq e + 2\delta.$$

Proof. Each good positive branch contributes at most $e \operatorname{Tr} \sigma_z$, by the diamond bound; each bad branch contributes at most $2 \operatorname{Tr} \sigma_z$, because both maps are channels. Sum. Later trace-preserving source/output operations leave the archived marginal unchanged by no-signalling. No independence assumption is used. □

Choose

$$T_0 = \kappa_0^{-1} \log(400n/\epsilon^2), \quad \eta = \min\{\Lambda/16, \epsilon/(100MT_0)\}, \quad \delta = \epsilon/100.$$

Capture the first input with P1's Hamiltonian-independent collectors, retaining flags and quantum exports. Learn H from later output by either construction. Clip estimates to supplied coefficient bounds, so bad estimates also specify finite channels. On every good record,

$$\left\|H - \widehat{H}\right\| \le M\eta, \qquad \left\|\mathcal{D}_{\widehat{H},\gamma} - \mathcal{D}_{H,\gamma}\right\|_\diamond \le 2T_0 M\eta$$

for every early history γ. Telescope receiver Hamiltonian intervals of total duration T_0; SWAPs and partial traces are contractions. Use a common finite decoder layout fixed by candidate supports and bounds, changing gate angles with the estimates. There is no extra export-count factor, and $\widehat{H}$ need not itself satisfy the extinction threshold.

Theorem 4 (First-payload recovery without prior Hamiltonian calibration)**.** *Under Section 2's premises, fixed k, and $r > \beta_0$, a finite output-only protocol recovers the first arbitrary input/reference with diamond error at most ϵ. Work, exposure, memory and finite control specifications are polynomial in the declared size, physical norm/rate values, inverse margin and inverse accuracy parameters, with standard precision factors. The receiver and memory error bounds below must actually be supplied.*

Proof. Combine P1, uniform calibration confidence and the adaptive-decoder lemma. The complete allowance is

$$\begin{aligned}\epsilon_{\text{total}} \le{}& 2\sqrt{n\mathrm{e}^{-\kappa_0 T_0}} + 2T_0 M\eta + 2\delta + \epsilon_{\text{mix}} + \epsilon_{\text{cal,phys}} \\ &+ \epsilon_{\text{capture,phys}} + \epsilon_{\text{queue}} + \epsilon_{\text{inverse}} + \epsilon_{\text{memory}} + \epsilon_{\text{loss}} + \epsilon_{\text{other}}. \end{aligned} \tag{8}$$

Give each of the eight latter allowances at most $\epsilon/12$. The first three sum to at most 0.14ϵ, so the total is below ϵ. Set $\epsilon_{\text{mix}} = 0$ in the no-burn-in construction. All caps and precision demands are fixed before consuming the input. Overflow or failed calibration returns a fixed state and is charged in this unconditional guarantee. □

8 Continuous fields and the complete finite execution bill

Reset histories above describe an acquired instrument, not source interventions. P1's full source-and-output comparison, over a window u with bins at most Δ, is

$$\epsilon_{\text{split}}(u) \le 2\chi_0 \Delta \sqrt{R_{\text{ph}} u + (R_{\text{ph}} u)^2}, \qquad \chi_0 \ge \max_i \sum_{X \ni i} \operatorname{osc}(h_X). \tag{9}$$

The comparison's Hamiltonian kicks are not performed on the source. Unknown coefficients are unnecessary for choosing its mesh. Uncollected ports may be traced out after this full comparison.

The finite record machines are compared with the binned acquired instrument by bounded CPTP Lie splitting, conservatively costing $2\Lambda^2 T_{\text{obs}}\Delta$, $T_{\text{obs}} \le 2N_{\text{cal}}h$. This is separate from (9). Both errors are summed for the whole capped protocol. No unconditional interface error is promoted to an error bound on every rare normalized branch.

Choose $\Delta \le h$ so that the sum of (9) over at most $2N_{\text{cal}}$ windows and the record-machine comparison each fit their portions of $\epsilon_{\text{cal,phys}}$. This requires polynomially small Δ. Divide each nonzero window into equal bins to avoid arbitrarily short remainders. The least positive Chebyshev duration is bounded below polynomially in h/D^2; round it within the stated timing budget and, if necessary, one quarter of that duration.

Put $K_{\max} = \lceil h/\Delta \rceil + 1$ and

$$B_{\text{cal}} \le N_{\text{cal}}(k+1)K_{\max}, \qquad z = \epsilon_{\text{cal,phys}}/(8B_{\text{cal}}).$$

Allocate at most z to every local collector and associated endpoint cycle. P1's constant-dimensional collector has sufficient specifications

$$\text{peak} \le 16\sqrt{r}/z, \qquad \text{slew} = O\left(\frac{\sqrt{r}}{z\Delta_b} + \frac{r^{3/2}}{z^4}\right), \qquad \text{mesh} = O(1 + r^2\Delta_b^2 z^{-5})$$

for a bin of width Δ_b. These demands grow with the task. This does not solve P1's separately developing fixed-hardware interface.

A separate bank per selected site/bin avoids instantaneous endpoint conversion or readout. After each trial, execute and pay endpoint gates, flag readout, Pauli readout and refresh, then reuse the banks. A two-bank implementation instead requires P1's explicit recycling latency inequality. Include known propagation delay and timing precision.

Resource	Charged quantity and access
Original inputs	One arbitrary unencoded payload; no later copy is requested.
Calibration trials	Every signed-outcome attempt in (6), or every martingale trial, including rejected reset events.
Source exposure	A conservative schedule pads each trial to $t_{\text{burn}} + 3h + t_{\text{service}}$, with real service and selection time. Include T_0, propagation and decoder waiting. Remove only burn-in in the bounded-neighborhood version.
Early quantum buffer	A sufficient P1 cap is $m_* = \lceil (\mathrm{e}-1)rT_0 + \log(20n/\epsilon_{\text{queue}})\rceil$ per site. Store $O(n(m_*+1))$ payload qubits and the paid timing/flag record. Expected exports nrT_0 alone are not a worst-case resource cap.
Calibration workspace	$O((k+1)K_{\max})$ collector banks per trial in the conservative interface, with calibrated constant-size references prepared at every use. Replace $k+1$ by $c+1$ for neighborhoods.
Interface work	All B_{cal} collectors, their finite meshes, endpoint operations, readouts, refreshes and precision. Low retained-output count does not remove this work.
Inverse receiver	Local simulation of $-\widehat{H}$ for time T_0, reverse SWAPs and traceouts. A product formula uses $O(M[C_{\widehat{H}}T_0^2/\epsilon_{\text{sim}} + nm_* + 1])$ local exponentials, with $C_{\widehat{H}} = O((\sum_a \lvert\widehat{\theta}_a\rvert)^2)$. Routing and finite gate accuracy are additional polynomial factors.
Protected storage	For provisioned Q_{mem} qubits through real T_{wall}, require $Q_{\text{mem}}T_{\text{wall}}\nu_{\text{mem}} \le \epsilon_{\text{memory}}$, or pay a specified fault-tolerant realization. The original archive survives all calibration and computation.
Loss and other noise	Sum full-protocol allowances in (8). P1's per-window loss and bounded hidden-noise/model-discrepancy bounds can be used. Fixed nonzero error per operation is not free as work grows.

These are polynomial claims in physical *values*, not merely rate bit lengths. No arbitrarily large coupling is fast-forwarded and no long storage wait is free. The rate, axes and purifier access remain independently calibrated premises. Learning H does not certify those premises from the same data.

9 Executed evidence and an adverse resource point

Two new calculations accompany the proof. The first simulates the ideal front/back reset instruments for a noncommuting two-qubit model, including transpose signs, outside resets and finite-window timing. It checks the derivative identity and the $4B_H w$ bound. Synthetic Bernoulli

observations are generated from calculated probabilities; no physical data are claimed. The model is

$$H = 0.37XI - 0.23IY + 0.41ZZ + 0.19XY - 0.17ZX, \qquad r = 4.$$

For one selected four-export early history of duration 0.35, the normalized Choi trace error is 2.65×10^{-8} with known H and 0.01083 after the illustrative noisy fit. The output difference is 0.01083, below the model-mismatch bound 0.1092. These are diagnostics for a selected ideal record, not a worst-case diamond benchmark or an observed clock history. The fit's two million accepted Bernoulli shots per coefficient are sampled by binomial generation; the enormous rejected-attempt cost was not physically executed.

The second calculation uses a triangular moment ODE for the signed output record and finite Chebyshev interpolation of calculated means. The mean evaluation uses a 48th-order Taylor expansion. At degree ten the largest error among the five nonzero terms is 7.40×10^{-8}. The absent XX term is about -3.16×10^{-24}. Its input probabilities are calculated rather than acquired; 70-digit interpolation arithmetic is disclosed.

Coefficient	True	Degree-ten reconstruction
XI	0.37	0.370000000062
IY	−0.23	−0.230000000098
ZZ	0.41	0.409999928939
XY	0.19	0.189999980270
ZX	−0.17	−0.169999926010

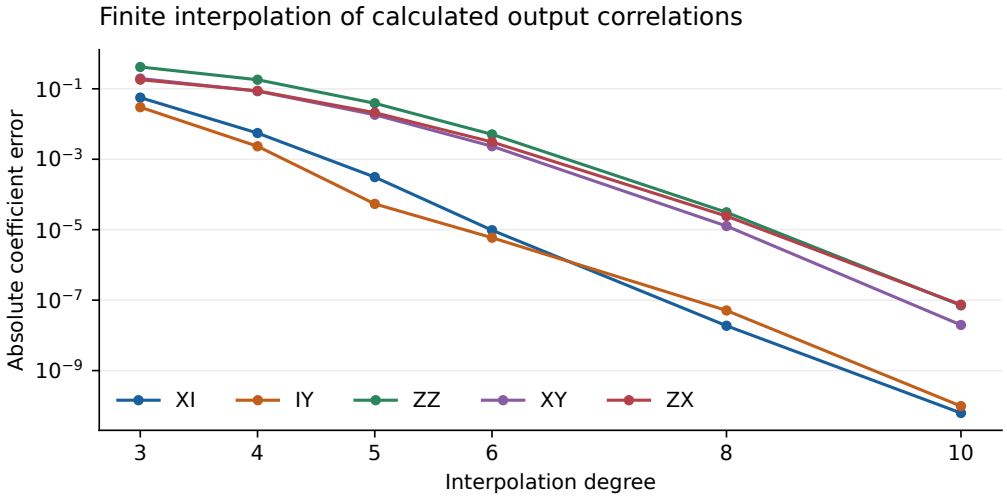


Figure 1: Executed finite interpolation of calculated ideal-instrument means. This checks the new acquisition identity on a small model. It is not experimental accuracy or an execution of the theorem's sampling budget.

The conservative theorem schedule was also evaluated. For $n = 2, M = 15, k = 2, B_H = 1.37, r = 4, \kappa_0 = 2.46, \epsilon = 0.1$, it gives

$$T_0 = 4.58934, \qquad \eta = 1.45264 \times 10^{-5}, \qquad h = 0.00667022.$$

Two fully specified sufficient schedules were evaluated:

Schedule	Degree	Total raw-trial upper bound
Norm/Markov, separate nodal batches	52	$10^{62.54}$
Factorial tail, actual weights, randomized nodes	14	$10^{39.12}$

The second evaluation inflates the calculated weight norm by a factor of two, uses the finite dyadic-sampler concentration bound, and requires timing accuracy about 9.35×10^{-24} in the model's time units. It applies with the two-site complete neighborhoods, so no burn-in is necessary. Both counts precede the continuous receiver compilation. These are upper bounds, not lower bounds or proposed device settings. It shows why a polynomial theorem alone does not establish a useful laboratory or civilisation-scale consequence. No theorem-scale sampling, protected factory or noisy field collector was simulated.

10 Resource boundaries and preserved construction knowledge

At source start I/d, discard every purifier A, retaining only P1's mixed exports B and their flags. For each reset history, source and fresh replacement baths are jointly maximally mixed, and every unitary segment and wire exchange preserves that joint state. The full retained B stream is therefore maximally mixed and independent of H. No collective measurement of this particular record can identify H. The positive operation uses additional actually delivered purifier information. This is not an impossibility claim for every classical measurement of the original fields.

If all payload information crosses an entanglement-breaking classical record and later calibration contains no surviving quantum payload, the final channel remains entanglement breaking. Its Choi overlap with Φ_d is at most $1/d$, so its distance from identity is at least $2(1 - 1/d)$. The early quantum archive is substantive.

Stationarity is substantive too: later observations cannot identify an arbitrary unobserved earlier drift. A changing Hamiltonian requires an independently bounded discrepancy allowance. Missing terms outside the candidate class, inaccessible source noise, unavailable purifiers and unknown port gauges are not inferred away.

The previous construction package is preserved byte-for-byte as a separate folder in the new source archive, including its complete proofs, code, evidence and limits. In particular:

1. The full classical-feedback all-strength hidden-Z theorem keeps its chronological full-CQ proof and per-branch budget. The sharper global-budget theorem is separate; its constant is not extended to arbitrary branchwise feedback.

2. The matched fixed-order known-parity/GHZ law, finite-slew thresholds and complete binary-POVM calibration remain. No general purity upper bound, growing-order sharp theorem or zero-capacity conclusion is added.

3. The independent thermal-effusion operation retains its certified thin barrier shell, actual force and clearing interface, finite angular quadrature, flight/clearing/calibration costs and residual exposure $1/(\Gamma\Lambda_{\mathrm{flux}})$. It is not general natural-failure prediction.

4. P1's known-Hamiltonian reader and separate diagonal-in-one-product-basis decoder keep their coherent-output and protected-memory premises.

5. P2's local unknown-support uncertainty construction stays within its declared local reach, product invariant law and operating refresh backbone. It does not certify arbitrary natural-system completeness.

The achieved new consequence is a change in experimental ordering: the first unknown state can be captured before its Hamiltonian is learned, with reference-uniform recovery and a complete finite interface. Practical efficiency and fixed physical hardware remain further requirements.

References

[1] Astra P1, *Scalable output-only acquisition from interacting many-body sources*, revision 324acb589443002bf8d529cc15e160129c5e36d2, https://github.com/harjassand/astra-research-knowledge/tree/324acb589443002bf8d529cc15e160129c5e36d2/research/returns/2026-11-05-p1-scalable-many-body-acquisition.

[2] Astra P3, *All-strength control speed*, same revision, https://github.com/harjassand/astra-research-knowledge/tree/324acb589443002bf8d529cc15e160129c5e36d2/research/returns/2026-11-05-p3-all-strength-control-speed.

[3] Astra P2, *Unknown local ports*, same revision, https://github.com/harjassand/astra-research-knowledge/tree/324acb589443002bf8d529cc15e160129c5e36d2/research/returns/2026-11-05-p2-unknown-local-ports.

[4] M. Guţă and J. Kiukas, *Equivalence classes and local asymptotic normality in system identification for quantum Markov chains*, Commun. Math. Phys. 335, 1397–1428 (2015), https://arxiv.org/abs/1402.3535.

[5] D. Stilck França, L. A. Markovich, V. V. Dobrovitski, A. H. Werner and J. Borregaard, *Efficient and robust estimation of many-qubit Hamiltonians*, Nature Communications 15, 311 (2024), https://doi.org/10.1038/s41467-023-44012-5.

[6] J. Castaneda and N. Wiebe, *Hamiltonian Learning via Shadow Tomography of Pseudo-Choi States*, Quantum 9, 1700 (2025), https://arxiv.org/abs/2308.13020.

[7] J. Haah, R. Kothari and E. Tang, *Learning quantum Hamiltonians from high-temperature Gibbs states and real-time evolutions*, Nature Physics 20, 1027–1031 (2024), https://arxiv.org/abs/2108.04842.

[8] W. Gong, M. Ma, S. Chen, J. Cotler and H.-Y. Huang, *Out-of-control Hamiltonian Learning*, arXiv:2610.06709 (5 October 2026), https://arxiv.org/abs/2610.06709.

[9] S. Liu, Z. Xu, A. V. Gorshkov, X. Wu, Y.-X. Wang and Z.-Y. Wei, *Learning Many-Body Hamiltonians Using a Local Probe*, arXiv:2610.02157 (1 October 2026), https://arxiv.org/abs/2610.02157.

[10] M. Navascués, *Resetting Uncontrolled Quantum Systems*, Phys. Rev. X 8, 031008 (2018), https://doi.org/10.1103/PhysRevX.8.031008.